SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 2 – Technical Presentation

Course Code:	DSA0311	**Course Title:**	Natural Language Processing
CO Assessed:	CO1 – Imitation (L1)	**Assessment:**	Assessment Tool 2 – Technical Presentation
Weightage:	30% of CIA		
CO1 Statement:	Analyse NLP models, regular expression patterns, finite state automata, morphological processing techniques and to interpret language processing. (BL-3)		
Student Name:	SATHISH R	**Reg. No.:**	192324204
Section / Batch:	B.Tech AI&DS	**Date:**	16-10-2026

Code of Conduct

I Sathish R 192324204 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Sathish R

Instructions

- Suggested Team Member Roles (3 Members)
- Member 1: Theory, Concepts, and Literature Review
- Member 2: Algorithm Design, Models, and Technical Analysis
- Member 3: Demonstration, Case Study, Applications, and Conclusion
- GPS photo must submit.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
Comparative Analysis of Inflectional and Derivational Morphology	Morphological Classification, Language Analysis	TP-10

Stemming vs Morphological Parsing

A Comparative Study of Word Reduction and Morphological Processing

1. Introduction

Stemming and morphological parsing are two important techniques in Natural Language Processing (NLP) used to analyze words. Stemming reduces words to a common stem by removing prefixes or suffixes, while morphological parsing analyzes the internal structure of words to identify roots, affixes, and grammatical features. For example, a stemmer may reduce studies to studi, whereas morphological parsing can identify the base word and interpret its grammatical role. This report explains both techniques, their differences, and their applications in NLP.

2. Objectives

To understand the concepts of stemming and morphological parsing.
To explain how stemming reduces words to their stems.
To analyze how morphological parsing identifies roots and affixes.
To compare the methods based on their processes, outputs, and complexity.
To understand the applications of both techniques in Natural Language Processing.
3. Stemming

Stemming is a word-reduction technique that removes common prefixes or suffixes to obtain a simplified form of a word. It generally uses simple rules or heuristics and does not always produce a valid dictionary word.

Examples:

Studies → Studi
Playing → Play
Connected → Connect
Happiness → Happi
Running → Run

Stemming is useful for grouping related words and reducing variations in text. It is commonly used in search engines, information retrieval, and text analysis.

4. Morphological Parsing

Morphological parsing is the process of analyzing the internal structure of a word to identify its root, affixes, and grammatical information. Unlike stemming, it aims to explain how a word is formed and what its components represent.

Examples:

Unhelpful → Un + Help + Ful
Teacher → Teach + Er
Rewriting → Re + Write + Ing
Happiness → Happy + Ness
Playing → Play + Ing

Morphological parsing can help identify grammatical features such as tense, number, and word class. The exact analysis may depend on the language and the context in which the word appears.

5. Comparison and Applications

Stemming and morphological parsing serve different purposes in NLP.

Aspect	**Stemming**	**Morphological Parsing**
Main purpose	Reduce words to stems	Analyze word structure
Output	Simplified stem	Root, affixes, and grammatical features
Complexity	Usually simpler and faster	Generally more complex
Linguistic detail	Limited	More detailed
Applications	Search engines and text indexing	Machine translation and linguistic analysis

Stemming is useful when the primary goal is to group related word forms. Morphological parsing is more appropriate when detailed information about word formation and grammar is required.

6. Presentation Image

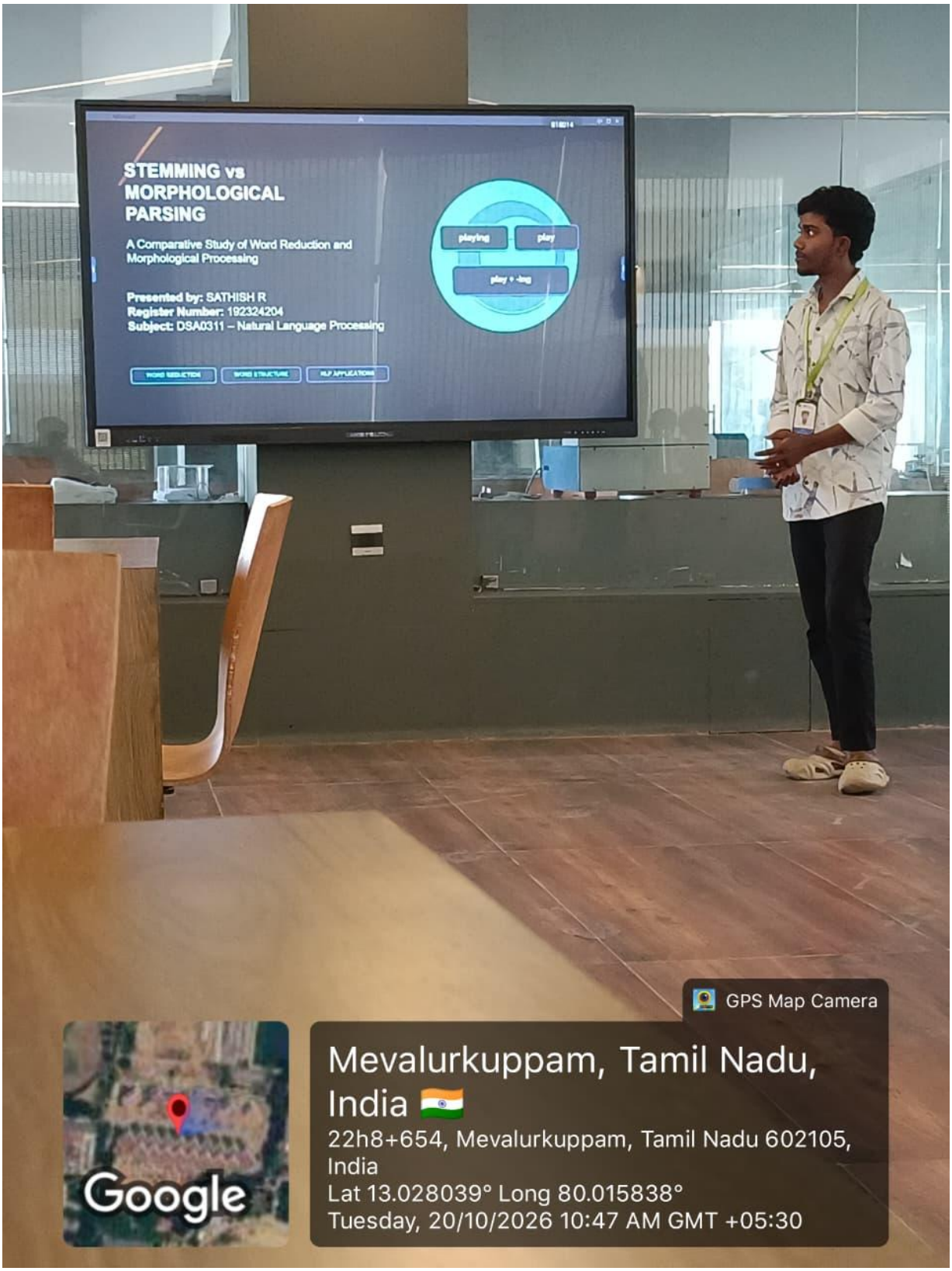


Figure 1: Slide from the presentation on Stemming vs Morphological Parsing.

7.Conclusion

Stemming and morphological parsing are important techniques in Natural Language Processing. Stemming simplifies words by reducing them to their stems, whereas morphological parsing identifies their internal structure and grammatical features. Understanding the differences between these methods helps determine which technique is most suitable for applications such as information retrieval, text processing, and machine translation.

Annexure A – Technical Presentation

Team No.	Technical Presentation Title	Description
1	Evolution of Natural Language Processing: From Rule-Based Systems to Modern NLP	Analyze the historical development of NLP models, key milestones, and how language processing has evolved from handcrafted rules to intelligent systems.
2	Comparative Study of NLP Models and Algorithms	Explore different NLP models and algorithms used for language processing, comparing their working principles, strengths, and limitations.
3	Regular Expressions for Real-World Text Processing	Demonstrate how regular expressions are used for data validation, text extraction, log analysis, and information retrieval in practical applications.
4	Designing Efficient Regular Expression Patterns for Information Extraction	Analyze complex regular expression patterns used to extract emails, phone numbers, URLs, dates, and structured information from unstructured text.
5	Finite State Automata in Natural Language Processing	Explain the concepts of deterministic and non-deterministic finite automata and their role in lexical analysis and language recognition.
6	Building a Finite State Automaton for Token Recognition	Design and demonstrate a finite state automaton that recognizes tokens such as identifiers, keywords, and numbers in textual data.
7	Morphology and Word Formation in Natural Languages	Investigate how words are structured and formed using morphemes, and analyze the importance of morphology in NLP systems.
8	Inflectional Morphology: Understanding Grammatical Variations	Examine how inflectional changes affect word forms based on tense, number, gender, and case across different languages.
9	Derivational Morphology and Vocabulary Expansion	Analyze derivational processes that create new words and study their impact on semantic interpretation and language understanding.
10	Comparative Analysis of Inflectional and Derivational Morphology	Compare the characteristics, applications, and challenges of inflectional and derivational morphology in NLP tasks.
11	Finite-State Morphological Parsing Techniques	Explore finite-state transducers and parsing techniques used to analyze and generate morphological structures in natural languages.
12	Morphological Analysis for Indian Languages	Study the challenges and solutions involved in performing morphological processing for morphologically rich Indian languages such as Tamil, Hindi, and Telugu.
13	Porter Stemmer: Design, Working Mechanism, and Applications	Analyze the Porter Stemming algorithm, its rule-based approach, implementation steps, and role in information retrieval systems.
14	Stemming vs Morphological Parsing: A Comparative Study	Compare stemming and morphological parsing techniques based on accuracy, computational complexity, and application scenarios.
15	Integrated NLP Pipeline: From Regular Expressions to Morphological Processing	Demonstrate a complete NLP preprocessing pipeline incorporating regular expressions, finite state automata, stemming, and morphological analysis for text processing.

Rubrics for Calculation

Technical Presentation					
Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism ; uneven team participation	Lacks professionalism ; minimal contribution or ethical awareness

Team No. / Criteria Level	Technical Content Accuracy	Problem Identification	Use of Tools	Communication	Professionalism	Total %

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____